Chapter 1

Exercise 1.2.A.

Proof. The two notions can be identified. Indeed, given a group G in the usual sense, define a category with a single object X, and for each $g \in G$, define a morphism $f_g : X \to X$. Further, for two morphisms f_g, f_h, define composition by $f_g \circ f_h = f_{gh}$. This gives a 1-object groupoid, since each morphism is an isomorphism: the morphism f_g has inverse $f_{g^{-1}}$.

Conversely, given a 1-object groupoid, the set of morphisms on that one object forms a group.

On the other hand, there are clearly groupoids that aren't groups, e.g. a category with two objects and only the identity morphisms on those objects. □

Exercise 1.2.B.

Proof. The identity morphism is invertible, with itself as left- and right-inverse. The composition of two invertible morphisms is again invertible, since $(f \circ g) \circ (g^{-1} \circ f^{-1}) = \mathrm{id}$. Finally, the inverse of an invertible morphism is invertible. So, the set of invertible morphisms forms a group under composition.

Given a set X, the automorphisms of X in the category of sets is the set of bijections $f : X \to X$, i.e. the automorphism group is S_X, the symmetric group on X. The automorphisms of a vector space in Vec_k are the automorphisms in the usual sense in linear algebra.

Finally, if A and B are isomorphic objects, with an isomorphism $f : A \to B$, then for an automorphism $\phi : A \to A$, the morphism $f \circ \phi \circ f^{-1}$ is an automorphism of B, with inverse $f \circ \phi^{-1} \circ f^{-1}$. So, this association gives an isomorphism of automorphism groups. □

Exercise 1.2.C. Skipped upon recommendation
Exercise 1.2.D. Skipped upon recommendation